21. Merge Two Sorted Lists

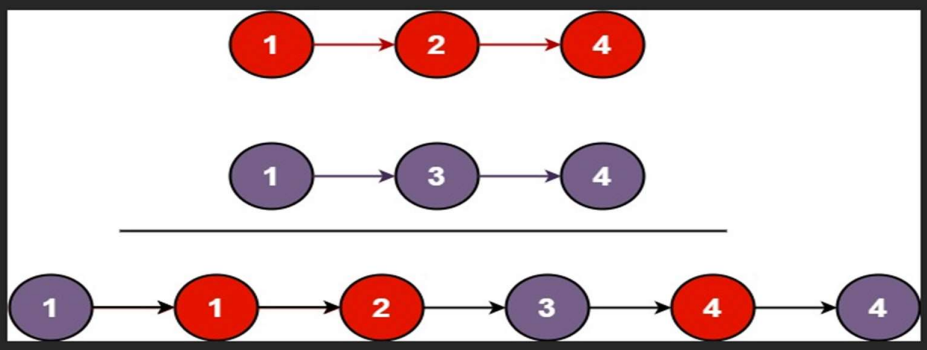
Easy | Topics | Companies

You are given the heads of two sorted linked lists `list1` and `list2`.

Merge the two lists into one **sorted** list. The list should be made by splicing together the nodes of the first two lists.

Return *the head of the merged linked list.*

Example 1:




```
Input: list1 = [1,2,4], list2 = [1,3,4]
Output: [1,1,2,3,4,4]
```

Example 2:

```
Input: list1 = [], list2 = []
Output: []
```

```c
/**
 * Definition for singly-linked list.
 * struct ListNode {
 * int val;
 * struct ListNode *next;
 * };
 */

struct ListNode* mergeTwoLists(struct ListNode* list1, struct ListNode* list2) {

    struct ListNode dummy; // dummy node
    struct ListNode* tail = &dummy; // pointer to build result

    dummy.next = NULL;

    while (list1 != NULL && list2 != NULL) {
        if (list1->val <= list2->val) {
            tail->next = list1;
            list1 = list1->next;
        } else {
            tail->next = list2;
            list2 = list2->next;
        }
        tail = tail->next;
    }

    // Attach remaining nodes
    if (list1 != NULL)
        tail->next = list1;
    else
        tail->next = list2;

    return dummy.next;
}
```

Accepted Runtime: 0 ms

Case 1 Case 2 Case 3

Input

list1 =

[1,2,4]

list2 =

[1,3,4]

Output

[1,1,2,3,4,4]

Expected

[1,1,2,3,4,4]

Accepted Runtime: 0 ms

Case 1 Case 2 Case 3

Input

list1 =

[]

list2 =

[]

Output

[]

Expected

[]

Accepted Runtime: 0 ms

Case 1 Case 2 Case 3

Input

list1 =

[]

list2 =

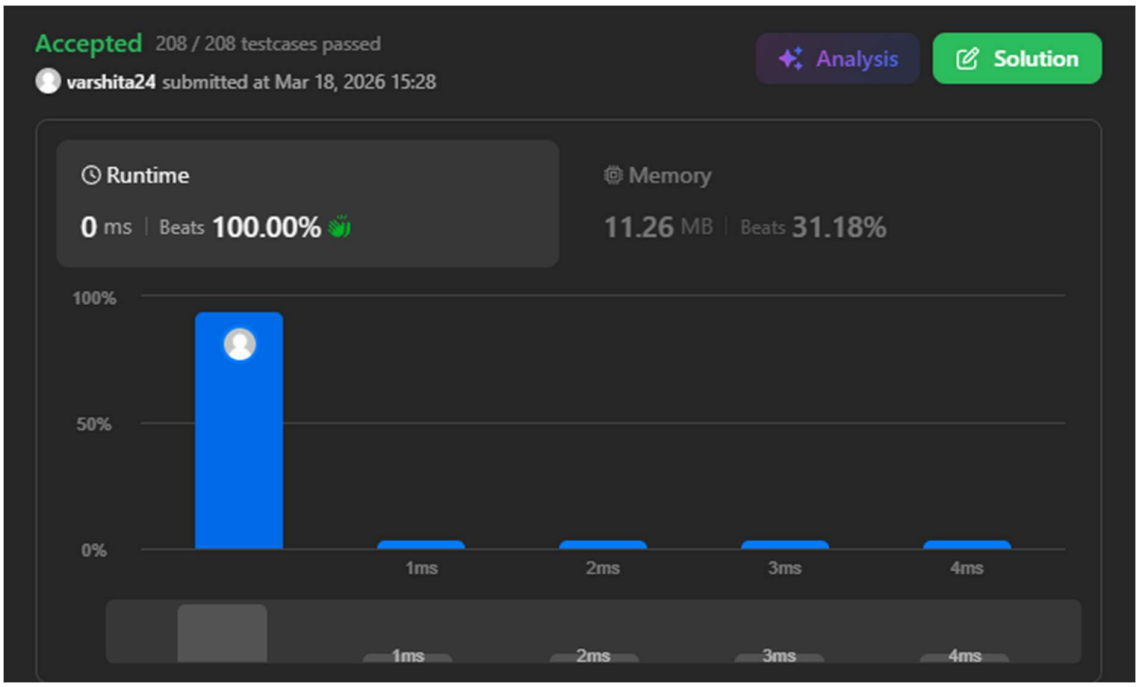
[0]

Output

[0]

Expected

[0]


Accepted 208 / 208 testcases passed
varshita24 submitted at Mar 18, 2026 15:28
Analysis
Solution
Runtime
0 ms | Beats 100.00%
Memory
11.26 MB | Beats 31.18%
100%
50%
0%
1ms
2ms
3ms
4ms